

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024

Data-forward brief for policy analysts and climate negotiators

Key Anchors

- Atmospheric CO₂: 429 ppm
- Annual emissions: >35 Bt
- China: ~12.5 Bt
- Per-capita gap: 150×

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

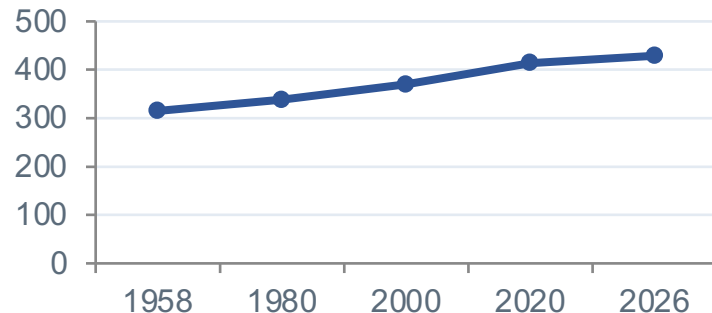
Keeling Curve context and pre-industrial baseline

429 ppm

Mauna Loa monthly concentration
February 2026

Pre-industrial baseline: ~280 ppm
Increase: +149 ppm (~+50%)

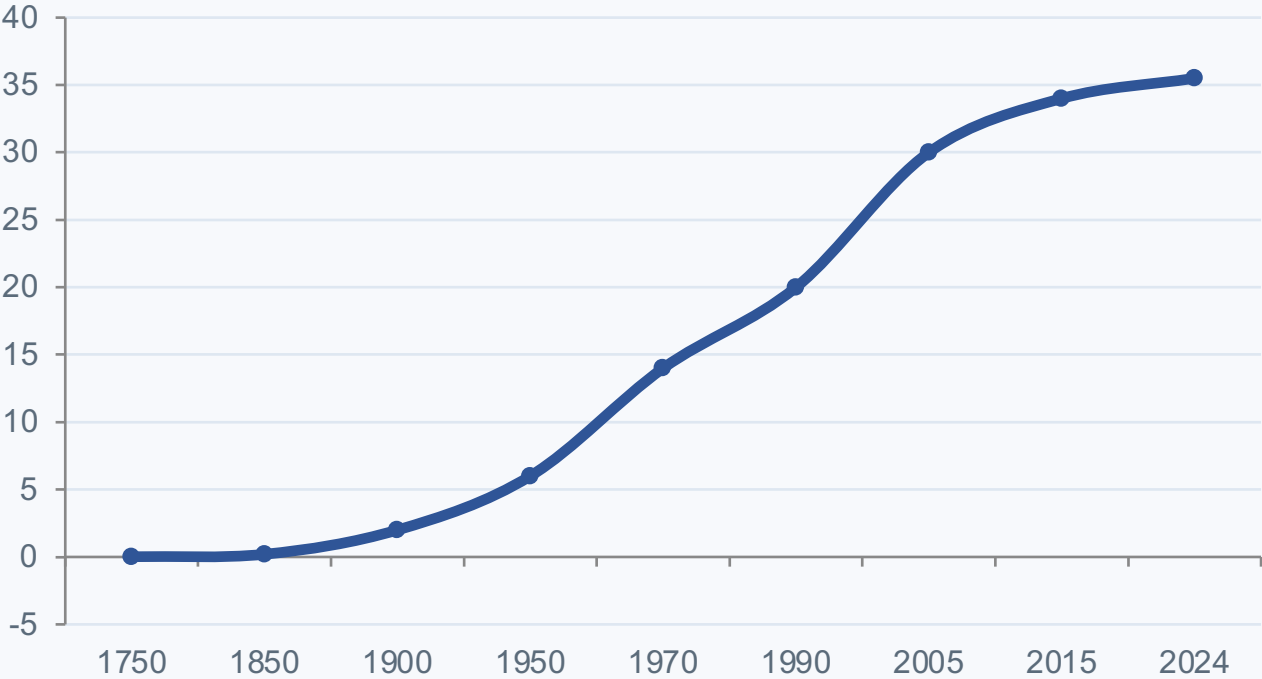
Acceleration Since 1958



Instrumental observations show persistent upward trend, with recent years reaching record highs.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

Historical trajectory from 1750 to 2024 (Figure 2)



Inflection Points

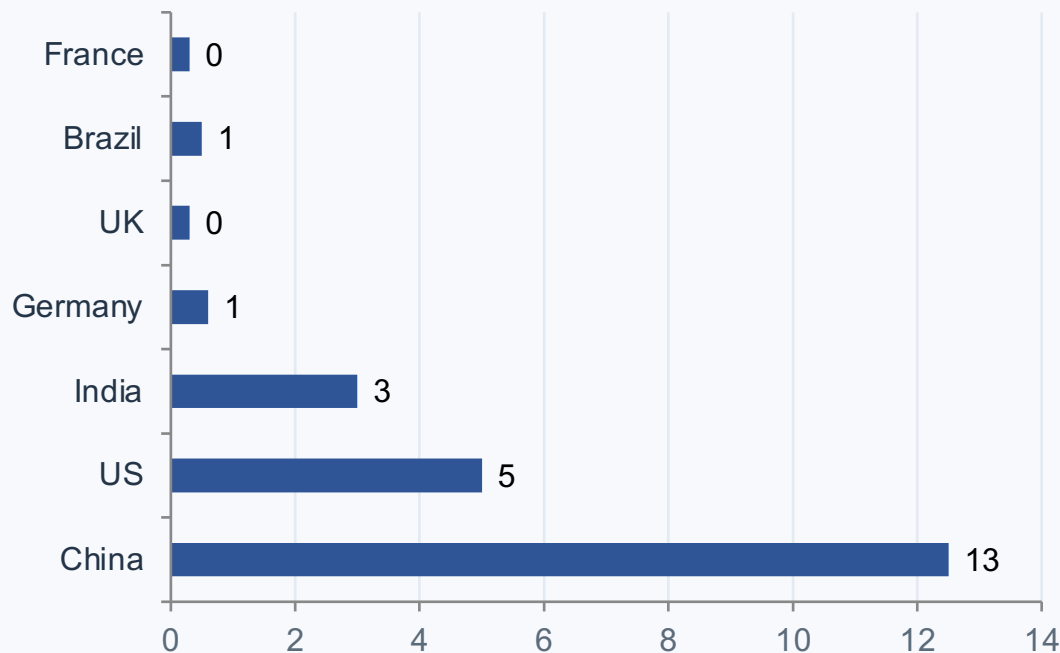
1950
6 Bt

1990
20 Bt

2024
35+ Bt

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

National annual CO₂ totals (2024) with historical context (Figure 3)

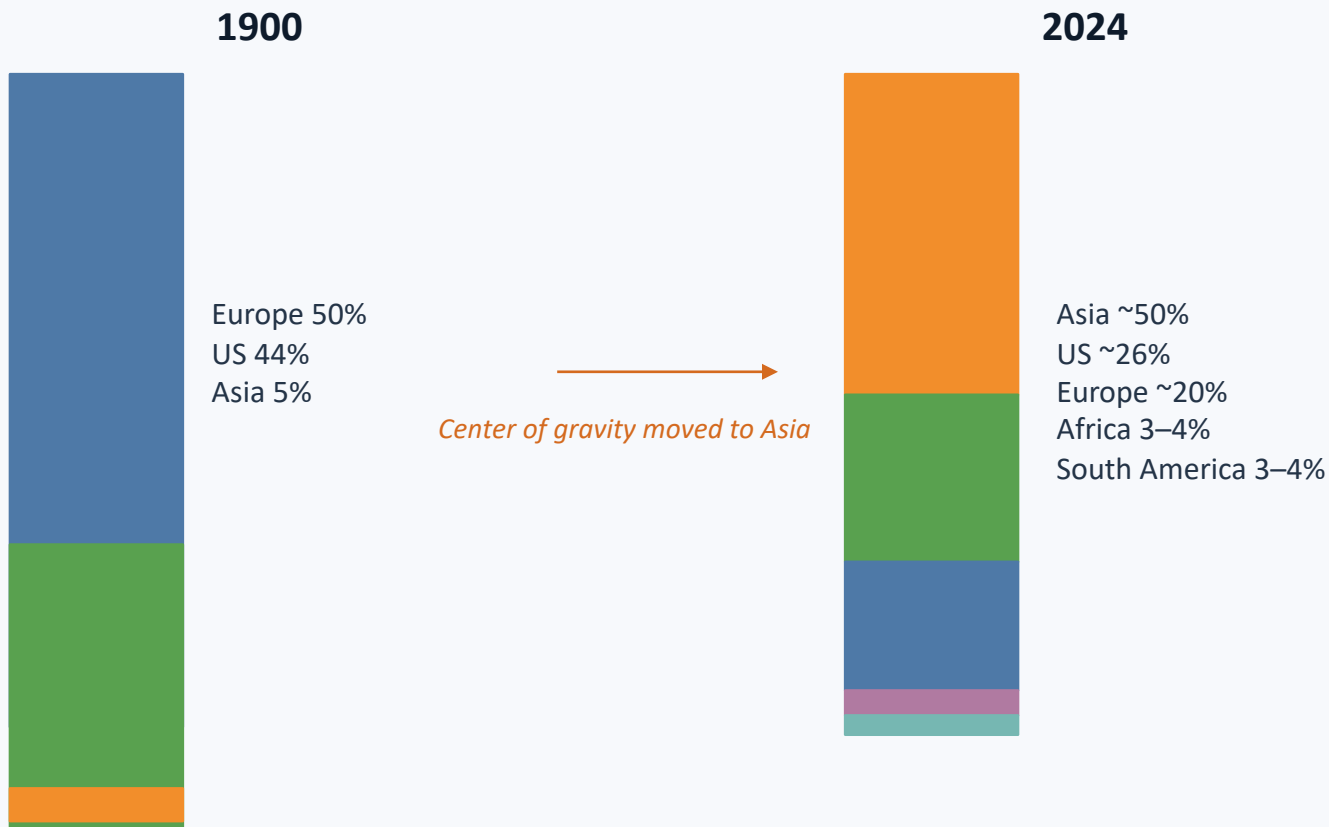


Interpretation

- China contributes >25% of global annual emissions.
- US emissions have declined from an earlier ~6 Bt peak to ~5 Bt.
- India is now the third-largest emitter and still rising rapidly.

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

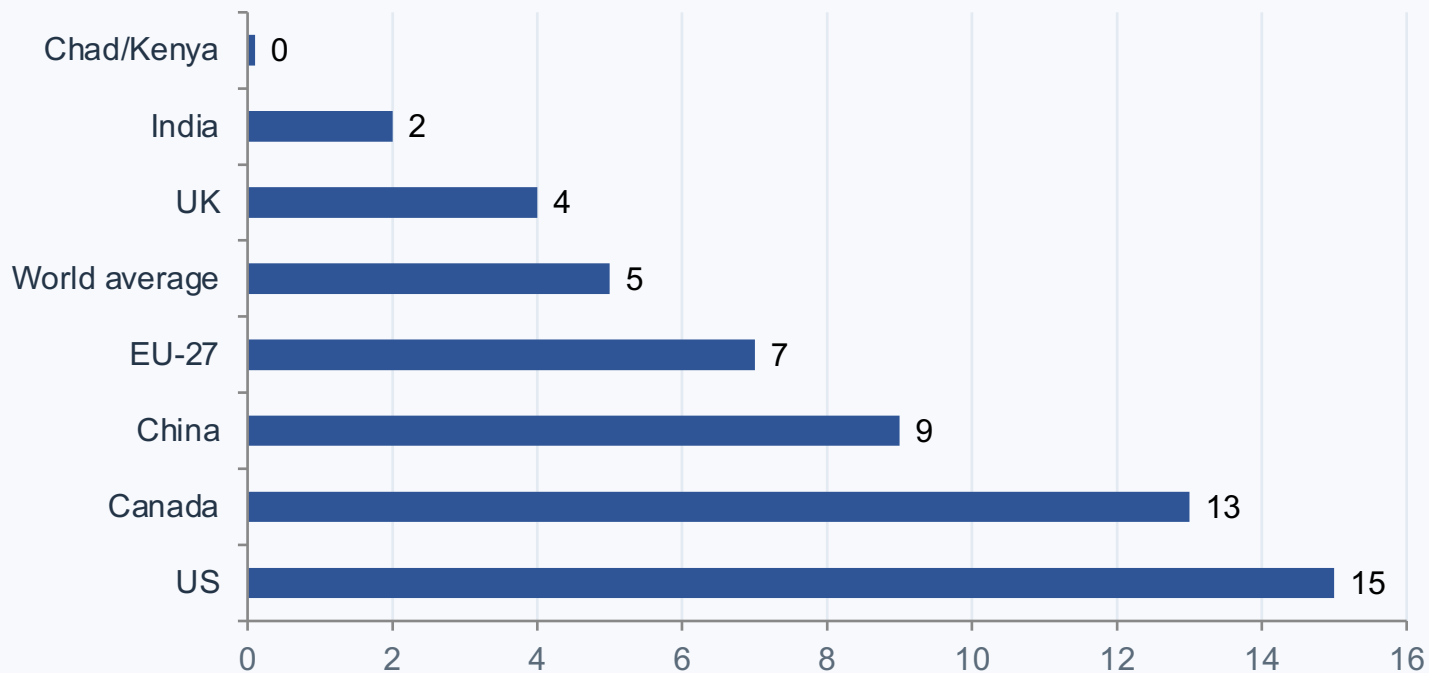
Regional share shift from 1900 to 2024



Source: source_brief.md regional shares (1900 vs 2024). Visual format follows requested side-by-side comparison.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

Horizontal comparison across selected economies (Figure 1 context)



$$15 \div 0.1 = 150\times$$

highest
vs lowest

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual percentage change map patterns (Figure 4)

Blue zones
-1% to -5%

Orange/red zones
+2% to +10%+

*Figure reference: `figure\_4.jpg`
(Annual % change map, 2024)*

Regional patterns

- US and much of Europe: declines (blue).
- Russia and parts of Southeast Asia: increases.
- Several African countries: +2% to +10%.
- Some South American countries: >10% increases; others declined.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Prosperity alone does not determine per-capita CO₂ outcomes

Countries with comparable living standards can diverge markedly in per-capita CO₂ depending on electricity and heat supply mix.

Country Pair	Dominant Power Mix (Qualitative)	Per Capita CO ₂ (t/person)
France	High nuclear + renewables; lower fossil share	4.5
UK	Growing renewables; reduced coal	4.0
Germany	Higher fossil-fuel share despite transition	7.0
Netherlands	Relatively higher fossil dependence	8.0

Policy implication: decarbonizing power supply composition (nuclear, renewables, storage, and grid reforms) can reduce per-capita emissions without requiring lower living standards.

Key Takeaways

- Global annual CO₂ emissions exceed 35 billion tonnes and have not yet peaked.
- China contributes over 25% of the global total, with annual emissions around 12.5 Bt.
- The emissions map has shifted: Asia is now ~50%, while US + Europe are below one-third.
- Per-capita inequality remains extreme: ~15 t/person in the US vs ~0.1 t/person in Chad (150×).
- Energy policy choices, especially power mix, strongly shape per-capita outcomes among advanced economies.